Homework 2 — changed numbers (extra practice)

Module 2 - IV Bolus Administration · Homework · No date

The numbers on this sheet are changed from her sheet.

Name _____ Date _____

1. A 176-lb male received a single IV bolus dose of an antibacterial agent at a dose level of 15 mg/kg. The concentration of drug in the plasma was 75.5 mcg/mL and 36.8 mcg/mL at 2 and 8 hours, respectively, following administration of the dose. The antibacterial agent displays, linear, first-order, one compartment pharmacokinetics.

1a What is the half-life of this agent in this patient?

1b What is the expected initial plasma drug concentration of this dose of drug in this patient?

1c What is the volume of distribution of this agent in this patient?

1d How many hours following administration of the dose are required for 87.5% of the agent to be eliminated from the body?

1e If the dose were administered at a level of 7.5 mg/kg, how many hours following administration of the dose are required for 87.5% of the drug to be eliminated from the body?

1f What is the clearance of this agent in this patient?

2. The equation below represents the concentration of drug in the plasma following IV bolus administration of a 350 mg bolus dose. (Concentration is given in mcg/mL and time in hours)

$C_p = 8.72e^{-3.85t} + 5.40e^{-0.154t}$

2a What is the elimination half-life of this agent?

2b What is the initial concentration of drug in the plasma upon administration of the dose?

2c What is the concentration of drug in the plasma 4 hours after administration of the dose?

2d What is the apparent volume of distribution of the central compartment?

Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working.

1a

Worked by the drill

Answer: 5.79 hr

WHAT THE STEM GIVES

weight	176-lb	*not needed: the half-life comes from the two concentrations*
D_0	15 mg/kg	*not needed*
C_1	75.5 mcg/mL at 2 hours	the numerator inside the logarithm
C_2	36.8 mcg/mL at 8 hours	the denominator inside the logarithm

THE WORKING, ONE LINE AT A TIME

SETUP

$$k = \frac{\ln(75.5/36.8)}{8\ \text{hr} - 2\ \text{hr}} = \frac{0.7186}{6\ \text{hr}} = 0.1198\ \text{hr}^{-1}$$

First order, so the rate constant is the log of the ratio of the two concentrations over the time between them.

ALGEBRA

$$t½ = \frac{0.693}{0.1198\ \text{hr}^{-1}} = 5.79\ \text{hr}$$

$t½ = \frac{0.693}{k}$, because $k\ t½ = \ln 2 = 0.693$ when the concentration falls to half. With $k = 0.1198\ \text{hr}^{-1}$ the reciprocal hours invert to hours: 5.79 hr per halving. Check: 2 to 8 hr is 1.04 half-lives, and the concentration fell from 75.5 to 36.8 mcg/mL, about half.

> **DOES THE NUMBER MAKE SENSE?**
> From 2 to 8 hr the level fell from 75.5 to 36.8 mcg/mL, almost exactly half, so the 6 hours between the samples is about one half-life: 5.79 hr, between 3 and 6.

Her homework wording with different numbers, so this is not the graded answer.

1b

Worked by the drill

Answer: 95.9 mcg/mL

WHAT THE STEM GIVES

weight	176-lb	*not needed: C_0 comes from extrapolating a measured level*
D_0	15 mg/kg	*not needed here; used with C_0 in part c*
C at 2 hours	75.5 mcg/mL	multiplied by e^{kt} to undo 2 hours of decline
C at 8 hours	36.8 mcg/mL	gave $k = 0.1198\ \text{hr}^{-1}$ in part a

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_0 = C\,e^{kt} = (75.5\ \text{mcg/mL})\,e^{(0.1198)(2)}$$

The one-compartment bolus line $C = C_0\,e^{-kt}$ rearranges to $C_0 = C\,e^{kt}$, where C_0 is the concentration the moment the dose is in. The first sample, 75.5 mcg/mL at 2 hr, has already declined for 2 hours at k = 0.1198 hr^{-1}; multiplying by e^{kt} undoes that decline.

ALGEBRA

$$C_0 = (75.5)(1.2707) = 95.9\ \text{mcg/mL}$$

kt = (0.1198 hr^{-1})(2 hr) = 0.2395, a pure number because the hours cancel, and $e^{0.2395}$ = 1.2707. (75.5 mcg/mL)(1.2707) = 95.9 mcg/mL: C_0 keeps the unit of the measured concentration and is higher than the 2 hr sample, as the level at time zero must be.

DOES THE NUMBER MAKE SENSE?

C_0 has to be above the 2-hour level of 75.5 mcg/mL, and 2 hours at k = 0.1198 hr^{-1} is about a third of a half-life, so the rise back is well under a doubling: 95.9 mcg/mL, between 75.5 and 2 × 75.5.

Her homework wording with different numbers, so this is not the graded answer.

1c

Worked by the drill

Answer: 12.51 L

WHAT THE STEM GIVES

weight	176-lb	converted: $\frac{176}{2.2} = 80.0$ kg
D_0	15 mg/kg	times 80.0 kg gives D_0 = 1200 mg, the numerator
C at 2 hours	75.5 mcg/mL	extrapolated in part b to C_0 = 95.9 mcg/mL, the denominator
C at 8 hours	36.8 mcg/mL	*not needed here: it served part a*

THE WORKING, ONE LINE AT A TIME

UNIT

$$176\ \frac{\text{lb}}{2.2}\ \text{lb/kg} = 80.0\ \text{kg};\ \text{dose} = (15\ \text{mg/kg})(80.0\ \text{kg}) = 1200\ \text{mg}$$

The dose is 15 mg for each kilogram of body weight, and the weight is in pounds; 2.2 lb is 1 kg, so $\frac{176\ \text{lb}}{2.2\ \text{lb/kg}} = 80.0$ kg, the pounds cancelling. (15 mg/kg)(80.0 kg) = 1200 mg: the kilograms cancel, leaving the whole dose D_0 in milligrams.

ALGEBRA

$$V_D = \frac{1200\ \text{mg}}{95.9\ \text{mg/L}} = 12.51\ \text{L}$$

Concentration is amount over volume, $C_0 = \frac{D_0}{V_D}$, so $V_D = \frac{D_0}{C_0}$: the volume of distribution is the apparent volume the 1200 mg dose occupies at 95.9 mcg/mL. 1 mcg/mL is 1 mg/L, so mg over mg/L cancels the milligrams and leaves litres, 12.51 L.

DOES THE NUMBER MAKE SENSE?

A 1200 mg dose giving 95.9 mg/L at time zero occupies $\frac{1200}{95.9}$, a little over 12 L: 12.51 L, above zero.

Her homework wording with different numbers, so this is not the graded answer.

1d

Worked by the drill

Answer: 17.36 hr

WHAT THE STEM GIVES

weight	176-lb	*not needed: the time to a fraction does not depend on the dose*
D_0	15 mg/kg	*not needed*
fraction eliminated	87.5%	leaves 12.5%, three half-lives
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave t½ = 5.79 hr in part a

THE WORKING, ONE LINE AT A TIME

SETUP

87.5% eliminated leaves 12.5%: 100 → 50 → 25 → 12.5, three half-lives

87.5% eliminated means 100 − 87.5 = 12.5% of the dose remains. With first-order elimination each half-life t½ removes half of what is left: 100% to 50% after one, to 25% after two, to 12.5% after three, so the time to 12.5% remaining is three half-lives.

ALGEBRA

t = 3 × 5.786 hr = 17.36 hr

3 × 5.786 hr = 17.36 hr. The same result comes from $C = C_0\, e^{-kt}$ with C over C_0 = 0.125: $t = \frac{\ln 8}{k} = \frac{2.0794}{0.1198\ \text{hr}^{-1}}$ = 17.36 hr. The time to a stated fraction eliminated depends on k alone, not on the dose.

DOES THE NUMBER MAKE SENSE?

87.5% gone leaves 12.5%, which is 100 → 50 → 25 → 12.5, three halvings, so the time is three half-lives: 3 × 5.786 = 17.36 hr, more than the 6 hours between the two samples.

Her homework wording with different numbers, so this is not the graded answer.

1e

Worked by the drill

Answer: 17.36 hr

WHAT THE STEM GIVES

D_0	7.5 mg/kg	*not needed: a first-order time to a fraction does not*

		depend on the dose
fraction eliminated	87.5%	leaves 12.5%, three half-lives
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave t½ = 5.79 hr in part a

THE WORKING, ONE LINE AT A TIME

SETUP

Halving the dose does not change k, so t = 3 × 5.786 hr = 17.36 hr

With first-order elimination the same fraction is lost per hour at any dose, so the time to lose 87.5% is unchanged.

> **DOES THE NUMBER MAKE SENSE?**
> Halving the dose halves every concentration but not the fraction lost per hour, so the time to lose 87.5% is still three half-lives: 17.36 hr, the same as part d.

Her homework wording with different numbers, so this is not the graded answer.

1f

Worked by the drill

Answer: 1.5 L/hr

WHAT THE STEM GIVES

weight	176-lb	with the dose gave V_D = 12.51 L in part c
D_0	15 mg/kg	with the weight gave the 1200 mg behind V_D
table	75.5 and 36.8 mcg/mL at 2 and 8 hours	gave k = 0.1198 hr^{-1} in part a

THE WORKING, ONE LINE AT A TIME

SETUP

$Cl = k \times V_D = (0.1198\ hr^{-1})(12.51\ L) = 1.50\ L/hr$

Clearance Cl is the volume of plasma cleared of drug per unit time; with first-order elimination the fraction k of the drug in the body leaves each hour, and that drug was spread through V_D, so $Cl = k \times V_D$. (0.1198 hr^{-1})(12.51 L) = 1.50 L/hr, reciprocal hours times litres.

> **DOES THE NUMBER MAKE SENSE?**
> The fraction k = 0.1198 of the 12.51 L leaves each hour, so about an eighth of the volume is cleared per hour: 1.50 L/hr, well below the 12.51 L volume.

Her homework wording with different numbers, so this is not the graded answer.

2a

Worked by the drill

Answer: 4.5 hr

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed: half-life does not depend on the dose*
A	8.72	*not needed: an intercept, not a rate*
α	3.85	*not needed: the larger exponent is distribution*
B	5.40	*not needed*
β	0.154	the denominator of $\frac{0.693}{\beta}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t½\beta = \frac{0.693}{\beta} = \frac{0.693}{0.154\ \text{hr}^{-1}} = 4.50\ \text{hr}$$

The elimination phase is the smaller exponent, β; the larger, α, is distribution and is not the half-life asked for.

> **DOES THE NUMBER MAKE SENSE?**
> β = 0.154 hr^{-1} removes about 15% an hour, so halving takes between 4 and 5 hours: 4.50 hr. The α term, with a half-life well under an hour, is gone long before.

Her homework wording with different numbers, so this is not the graded answer.

2b

Worked by the drill

Answer: 14.12 mcg/mL

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed: the intercepts already carry the dose*
A	8.72	added to B
α	3.85	*not needed: at t = 0 both exponentials equal 1*
B	5.40	added to A
β	0.154	*not needed: at t = 0 both exponentials equal 1*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p^{\,0} = A + B = 8.72 + 5.40 = 14.12\ \text{mcg/mL}$$

$C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$ gives the concentration at any time; at t = 0 both exponents are zero and $e^0 = 1$, so each term equals its intercept and $C_p^{\,0} = A + B = 8.72 + 5.40 = 14.12$ mcg/mL, the concentration the moment the 350 mg bolus is in.

> **DOES THE NUMBER MAKE SENSE?**
> At time zero both exponentials are 1, so $C_p^{\,0}$ is the plain sum 8.72 + 5.40 = 14.12 mcg/mL, above either intercept alone.

Her homework wording with different numbers, so this is not the graded answer.

2c Worked by the drill

Answer: 2.92 mcg/mL

WHAT THE STEM GIVES

D_0	350 mg bolus dose	*not needed*
A	8.72	the coefficient of $e^{-\alpha t}$
α	3.85	the exponent $\alpha t = 3.85 \times 4 = 15.4$
B	5.40	the coefficient of $e^{-\beta t}$
β	0.154	the exponent $\beta t = 0.154 \times 4 = 0.616$
t	4 hours	t in both exponentials

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_p = 8.72e^{-(3.85)(4)} + 5.40e^{-(0.154)(4)} = 8.72e^{-15.4} + 5.40e^{-0.616}$$

The two-compartment equation $C_p = A\,e^{-\alpha t} + B\,e^{-\beta t}$ holds at any time, so both terms are evaluated at t = 4 hr. α = 3.85 hr^{-1} is the distribution rate constant and β = 0.154 hr^{-1} the elimination rate constant; each exponent is hr^{-1} times hours, a pure number: 15.4 and 0.616.

ALGEBRA

$$C_p = 0.00000 + 2.917 = 2.92 \text{ mcg/mL}$$

By 4 hours the distribution term is almost gone, so the elimination term carries nearly all the concentration.

> **DOES THE NUMBER MAKE SENSE?**
>
> By 4 hours the α term, $e^{-15.4}$, is zero to five decimals, so only the B term is left; βt = 0.616 is a bit under 0.693, so a bit over half of 5.40 remains: 2.92 mcg/mL, below 5.40.

Her homework wording with different numbers, so this is not the graded answer.

2d Worked by the drill

Answer: 24.79 L

WHAT THE STEM GIVES

D_0	350 mg bolus dose	the numerator
A	8.72	added to B for $C_p{}^0$
α	3.85	*not needed*
B	5.40	added to A for $C_p{}^0$ = 14.12 mg/L in part b, the denominator
β	0.154	*not needed*

THE WORKING, ONE LINE AT A TIME

SETUP

$$V_p = \frac{D_0}{A + B} = \frac{350\ \text{mg}}{14.12\ \text{mg/L}} = 24.79\ \text{L}$$

The central compartment is the volume the dose first distributes into, so $V_p = \frac{D_0}{C_p{}^0}$ with $C_p{}^0 = A + B = 14.12$ mcg/mL, the concentration at time zero. 1 mcg/mL equals 1 mg/L, so 350 mg over 14.12 mg/L cancels the milligrams and leaves litres, 24.79 L.

DOES THE NUMBER MAKE SENSE?

350 mg producing 14.12 mg/L at time zero fills $\frac{350}{14.12}$, about 25 L: 24.79 L, above zero.

Her homework wording with different numbers, so this is not the graded answer.